

Homework #3

Part 1: Instrument Variables - 1

The Excel file “*soft drinks.xls*” is an IRI dataset containing the information of aggregate store sales, prices and in-store promotion activities, including product displays and feature advertising, in the soft drinks category. It covers four major brands (Pepsi, Coca-Cola, Royal Crown and a “Private Label”) in seven stores in a mid-west city. (<https://pubsonline.informs.org/doi/pdf/10.1287/mksc.1080.0450>) Use the data to address the following questions:

1. Suppose you want to run a sales regression model. In particular, you are interested to understand the price promotion elasticity of demand. It is argued that own-brand price may correlate with the demand shock, i.e., the error term in regression. For example, there may be some promotional activities unobserved to researchers such as coupon that can change demand and manufacturers may coordinate such activities with price promotions. In this case the OLS regression estimators might be biased. A common practice is to use lagged prices (i.e. prices in past weeks. Columns 45-48 in data.) as instrumental variable (IV) for current prices to solve such endogeneity issues. Use your intuition to discuss why lagged prices may or may not be valid instruments for the soft drinks category.
2. Run the sales regression model using the OLS and IV methods, respectively. You can follow the R codes I provided below or use your own codes. In the IV method, your set of instruments should include the covariates in the data except own price, plus the four lagged prices (columns 45-48 in data.) Compare your results with that obtained from OLS estimation. Do you think the results are reasonable?

3. Based on the estimation results, run a statistical test to test if the explanatory variables in the model are exogenous. Further, test whether or not the lagged prices in your model are valid instruments. Discuss your results.

(1) Through the Hausman-Durbin-Wu Test, we find the test statistic $T = ?$, and $\Pr(\chi^2(1) \geq 3.84) = 0.05$. If $T > 3.84$, we could reject the null hypothesis that the explanatory variables in the model are exogenous at 5% level. Use the estimation results to draw conclusion accordingly.

(2) Through over-identification test, we find the test statistic $T1 = ?$, and $\Pr(\chi^2(3) \geq 7.81) = 0.05$. If $T1 > 7.81$, we could reject the null hypothesis that the instruments from question 2 (including lagged prices) are valid at 5% level. In addition, we may also only use the lagged prices as IV to do a 2SLS estimation. The results will show that the variance of estimators is very large since $(X'W(W'W)^{-1}W'X)$ is near singular? From this perspective, we could also verify that lagged prices are good instrument variable or not?

Based on the results of above two tests, draw your conclusion on the endogeneity problem here, and whether lagged prices are valid instruments or not?

R code for Part 1

```
library(readxl)
library(dplyr)
setwd("~/Data/Homework/HW3")
d<-read_xls("working data.xls")
N=nrow(d)
X=data.matrix(select(d, 2:28, 30:44 ))
X=cbind(X,intercept=1)
Y=data.matrix(select(d,`log(Sales)`))

####Q2 OLS
ols=solve(t(X)%*%X)%*%(t(X)%*%Y)
ols
error_ols=Y-X%*%ols
sample_error_ols=sum(error_ols^2)/(N-43)
cov_ols=sample_error_ols*(solve(t(X)%*%X))
sd_error_ols=sqrt(diag(cov_ols))
sd_error_ols
t_ols=ols/sd_error_ols
p_ols=2*pt(-abs(t_ols),df=N-43)
p_ols

###Q2 2SLS all exo var and lagged prices as iv
##stage 1
E=data.matrix(select(d,`own log(price/ medianprice)`))
W=data.matrix(select(d,2:28, 30:43, 45:48))
W=cbind(W,intercept=1)
iv_1=solve(t(W)%*%W)%*%(t(W)%*%E)
fitted_E=W%*%iv_1
X_W=cbind(X[,c(1:41)],fitted_E)
X_W=cbind(X_W,intercept=1)
```

##stage 2

iv_2=solve(t(X_W)%%X_W)%(t(X_W)%%Y)

iv_2

error=Y-X_W%%iv_2

sample_error=sum(error^2)/(N-43)

sd_error=sqrt(diag(sample_error*solve(t(X_W)%%W%%solve(t(W)%%W)%(t(W)%%X_W)))

sd_error

t=iv_2/sd_error

p=2*pt(-abs(t),df=N-43)

p

####Q2 IV method

P=W%%solve(t(W)%%W)%(t(W)

iv_ivm=solve(t(X)%%P%%X)%(t(X)%%P%%Y

iv_ivm

error_ivm=Y-X%%iv_ivm

sample_error_ivm=sum(error_ivm^2)/(N-43)

cov_ivm=sample_error_ivm*solve(t(X)%%P%%X)

sd_error_ivm=sqrt(diag(cov_ivm))

sd_error_ivm

t_ivm=iv_ivm/sd_error_ivm

p_ivm=2*pt(-abs(t_ivm),df=N-43)

p_ivm

####Q3 H-D-W Test

T=t(ols-iv_ivm)%%solve(cov_ivm-cov_ols)%(ols-iv_ivm)

T

c=qchisq(.95, df=1)

c

####Q3 Over-identification Test

v_hat=t(error_ivm)%%P%%error_ivm

T1=v_hat/sample_error_ivm

T1

```
c1=qchisq(.95, df=3)
```

```
c1
```

```
###2SLS only lagged price as iv to check if lagged prices are weak iv
```

```
##stage 1
```

```
E=data.matrix(select(d,`own log(price/ medianprice)`))
```

```
W3=data.matrix(select(d,45:48))
```

```
W3=cbind(W3,intercept=1)
```

```
iv_13=solve(t(W3)%*%W3)%*%(t(W3)%*%E)
```

```
fitted_E3=W3)%*%iv_13
```

```
X_W3=cbind(X[,c(1:41)],fitted_E3)
```

```
X_W3=cbind(X_W3,intercept=1)
```

```
##stage 2
```

```
iv_23=solve(t(X_W3)%*%X_W3)%*%(t(X_W3)%*%Y)
```

```
iv_23
```

```
error3=Y-X_W3)%*%iv_23
```

```
sample_error3=sum(error3^2)/(N-43)
```

```
solve(t(X_W3)%*%W3)%*%solve(t(W3)%*%W3)%*%(t(W3)%*%X_W3)
```

```
####the above matrix is singular, and therefore lagged prices are weak iv
```

```
#sd_error3=sqrt(diag(sample_error3*solve(t(X_W3)%*%W3)%*%solve(t(W3)%*%W3)%*%(t(W3)%*%  
X_W3)))
```

Part 2: Instrument Variables - 2

Now instead of writing our own codes, we use the R package to do the following. This is to show we can save some time using the existing package while learning how to do it in the previous question. The data set for this problem set contains simulated data. It is called *HW3_IV.xlsx*.

It contains the following variables:

y = dependent variable
x1 = exogenous regressor
x2 = endogenous regressor
z1 = potential instrument
z2 = potential instrument
z4 = potential instrument

2.1. Read in the data for the problem set and summarize it.

2.2. Regress y on x1 and x2. What is the estimated coefficient on x2?

2.3. Using the ivreg package in R, estimate the model in Problem 2.2 using z1 as an instrument. Describe the results and contrast them to those obtained in Problem 2.2 using OLS. <https://cran.r-project.org/web/packages/ivreg/vignettes/ivreg.html>

2.4. Perform a Durbin-Wu-Hausman test of the null that x2 is exogenous using z1 as the instrument. Can you reject the null? Does this or does this not surprise you in light of your answer to Problem 2.3?

<https://cran.r-project.org/web/packages/ivreg/vignettes/Diagnostics-for-2SLS-Regression.html>

2.5. Using the ivreg package, regress y on x1 and x2 using z1 and z2 as instruments. Describe your results. How do they differ, if at all, from those obtained in Problem 3? How would you expect them to differ?

Part 3: Manager's Pay – Panel Data

“The conflict of interest between shareholders of a publicly owned corporation and the corporation's manager is a classic example of a principal-agent problem... Managerial actions and investment opportunities are not perfectly observable by shareholders; indeed, shareholders do not often know what actions the manager can take or which of these actions will increase shareholders' benefits. In these situations, agency theory predicts that compensation policy will be designed to give the manager incentives to select and implement actions that increase shareholder wealth. Shareholders want managers to take particular actions – for example, deciding which issue to work on, which project to pursue, and which to drop – whenever the expected return on the action exceeds the expected costs. But the manager compares only his private gain and cost from pursuing a particular activity... Compensation policy that ties the manager's welfare to shareholder wealth helps align the private and social costs and benefits of alternative actions and thus provides incentives for managers to take appropriate actions... There are many mechanisms through which compensation policy can provide value-increasing incentives, including performance- based bonuses and salary revisions, stock options, and performance- based dismissal decisions.”

Agency theory predicts that compensation policy will tie the agent's expected utility to the principal's objective. Specifically, the formal model states that under an optimal contract the *pay-performance sensitivity* (when the manager is risk neutral) should be exactly equal to 1. That is, the manager receives the entire output as compensation but pays the shareholders an up-front fee. In this case the manager will only select an action that maximizes the company's wealth; hence, the principal-agent problem will be solved. Any contract with pay-performance sensitivity smaller than one will create conflicts in interest.

The purpose of this exercise is to understand the magnitude that manager's compensation relates to shareholders' wealth, results of which can be used to test the agency theory. The data in “*ManagerPay.xls*” are annual observations of the *Fortune 1,500* firms over the period 2004-07 (after excluding firms without complete observations in the four years. Hence, some small firms are excluded from the original dataset), taken from 10(k) filings and company reports.

To test the agency theory predictions, suppose the following is your regression model:

$$CEO_Pay_{i,t} = \alpha_0 + \lambda_1 \cdot Year_{2005} + \lambda_2 \cdot Year_{2006} + \lambda_3 \cdot Year_{2007} + \gamma \cdot Mkt_Value_{i,t} + e_{it} \quad (1)$$

where the variable “ $Year_k$ ” is an indicator that equals to 1 if $t = k$, and 0 otherwise, “ $CEO_Pay_{i,t}$ ” is the total manager compensation from firm i in year t , and “ $Mkt_Value_{i,t}$ ” is the total shareholders' wealth (end-of-year stock price multiplied by total number of shares).

1. Estimate equation (1) using both fixed-effects model and random-effects model approaches following R sample codes posted in Lecture 9. Discuss the model assumptions behind the two approaches.
2. Compare your results from the two approaches. Test whether the model assumptions related to the random-effects model are valid or not. (Hint: Hausman Test in the R sample codes posted in Lecture 9)
3. Based on your estimation results from fixed-effects and random-effects models, discuss whether the predictions from the agency theory are consistent with the data or not.

3180 observations, 55 columns.

The variables included:

Cols 1-4: Year indicator (2004-2007);

Cols 5-49: SIC 2-digit indicators;

Col 50: Firm number;

Col 51: CEO Bonus;

Col 52: CEO options;

Col 53: CEO salary;

Col 54: CEO total compensation;

Col 55: Market value

Sample R Codes:

```
library(readxl)
library(dplyr)
library(plm)
setwd("~/HW3")
d<-read_xls("ManagerPay.xls")

### calculate manually
Y=cbind(d$`CEO Total Compensation ($)`)
mv=d$`Market Value ($1,000)`/10^7
X=cbind(d$X__1, d$X__2, d$X__3, mv)
X1=cbind(intercept=1, d$X__1, d$X__2, d$X__3,mv)
##fixed effect model
T=4
K=4
e=matrix(1,nrow=T,ncol=1)
e1=matrix(1/T,nrow=T,ncol=1)
phi=1/T*(e%*%t(e))
N=nrow(d)/T
P=kronecker(diag(N),phi)
Q=diag((N*T))-P
```



```

beta_fixed=solve(t(X)%*%Q%*%X)%*%t(X)%*%Q%*%Y
beta_fixed
error_fixed=Q%*%Y-Q%*%X%*%beta_fixed
sample_error_fixed=sum(error_fixed^2)/(N*T-N-K)
var_fixed=sample_error_fixed*(solve(t(X)%*%Q%*%X))
se_fixed=sqrt(diag(var_fixed))
se_fixed
t_fixed=beta_fixed/se_fixed
p_fixed=2*pt(-abs(t_fixed),df=N*T-N-K)
p_fixed

##random effect model
P1=kronecker(diag(N),t(e1))
beta_1=solve(t(P1%*%mv)%*%(P1%*%mv))%*%t(P1%*%mv)%*%(P1%*%Y)
ksi=P1%*%Y-P1%*%mv%*%beta_1
sample_error1=T/(N-1)*t(ksi)%*%ksi
V_inv=(1/sample_error_fixed)*(diag(T)-(as.numeric((sample_error1-
sample_error_fixed)/T/sample_error1))*(e%*%t(e)))
V_inv
P2=kronecker(diag(N),V_inv)
beta_random=solve(t(X1)%*%P2%*%X1)%*%t(X1)%*%P2%*%Y
beta_random
var_random=solve(t(X1)%*%P2%*%X1)
se_random=sqrt(diag(var_random))
se_random

##Hausman's Test
Test=t(beta_fixed-beta_random[2:5])%*%solve(var_fixed-var_random[2:5,2:5])%*%(beta_fixed-
beta_random[2:5])
c=qchisq(.95, df=1)
c=qchisq(.99, df=1)
c

###use package
pdata<-pdata.frame(d,index = "Firm Number")
Y=cbind(d$`CEO Total Compensation ($)`
X=cbind(d$X__1, d$X__2, d$X__3, d$`Market Value ($1,000)`
##fixed effect model
fixed<-plm(Y~X,data = pdata, model="within")
summary(fixed)
##random effect model
random<-plm(Y~X,data=pdata, model="random")
summary(random)

##Hausman test
phtest(fixed, random)
##adjust by df=1

```

Part 4: Multinomial Choice Models

This problem set uses data from the National JTPA Study (NJS). JTPA stands for Job Training Partnership Act. Until 2000, when the Workforce Investment Act replaced it, the programs funded under JTPA constituted the largest federal training program in the United States. The NJS was an experimental evaluation of the JTPA program in the late 1980s and early 1990s.

Persons randomly assigned to the experimental treatment group during the NJS, and thus granted access to JTPA, could participate in one of several different types of employment and training services, or could chose to drop out and not participate at all. This problem set considers the determinants of the type of services that an individual receives conditional on receiving some service. Put differently, this problem set ignores the dropout decision.

Data Set: NJS Experimental Sample

The data consist of a random sample of 1500 of the adult (over age 21) female members of the experimental treatment group who received at least one service from JTPA during the 18 months after random assignment. Individuals receiving more than one service are coded based on the service with the most hours. The data appear in the file: *hw_multi.xlsx*.

The variables contained in the data set are defined as follows:

BIFID: Identification number (BIF = Background Information Form)

AGE: Age in years

STYPE: Service type received under JTPA

1 = Classroom training in occupational skills (CT)

2 = Subsidized on-the-job training at private firms (OJT)

3 = Job search assistance (JSA)

4 = Other

ED1: Dummy variable for < 12 years of schooling

ED2: Dummy variable for exactly 12 years of schooling

ED3: Dummy variable for > 12 years of schooling

BLACK: Dummy variable for Black

HISP: Dummy variable for Hispanic

NVRWRK: Dummy variable for never having worked for pay at the time of random assignment.

1. Using the data from *hw_multi.xlsx*, estimate a multinomial logit model with service type as the dependent variable and race/ethnicity, age, education and having never worked as independent variables. Use `mlogit` in R or other codes you feel comfortable. Interpret your results.

2. Estimate a multinomial probit model of service type using the same covariates as in Problem 1 and the `mprobit` command in R, which assumes independence of the random components of utility across services. Interpret your results.

3. Compare the various multinomial choice models in this context. Suppose that you were going to write a paper on the determinants of service choice using these data. Which model's results would you report and why?